

Charotar University of Science and Technology
M.Sc. (Nano Science & Technology) Semester I Examination Dec. 2009

NT701 Concepts of Physical Sciences I

Date: 01 December 2009

Time: 10.00 am to 10.30 pm

Max. Marks 20

Part – A

Do as directed (Show your calculation within the space provided)

Que.1 The c/a ratio for hexagonal closed packed structure is: (Tick the correct answer) (1)

- (i) $8/3$ (ii) $3/8$ (iii) $\sqrt{8/3}$ (iv) $\sqrt{3/8}$

Que.2 The density of iron is 7.86 g/cc and its atomic weight is 55.85. The atomic radius of iron having the BCC structure is : (Show the calculation and tick the correct answer) (3)

- (i) $2.42\text{\AA}$ (ii) $4.42\text{\AA}$ (iii) $1.42\text{\AA}$ (iv) $3.42\text{\AA}$

Que.3 The energy of photon of wavelength $2\text{\AA}$ in electron volt is: (show the calculation and tick the correct answer) (3)

- (i) 6198 eV (ii) 1968 eV (iii) 9681 eV (iv) 8961 eV

Que.4 The density of state having energy E is a function of: (Tick the correct one) (1)

- (i) E^2 (ii) $E^{1/2}$ (iii) E^3 (iv) $1/E$

Que.5 For an intrinsic semiconductor the conductivity varies with temperature: (1)

- (i) Inversely (ii) linearly (iii) Exponentially (iv) does not depend

Que.6 Indicate change or no change when metal transit from normal state to superconducting state: (4)

- (i) Crystal structure:
(ii) Resistance:
(iii) Optical reflectivity
(iv) Specific heat

Que.7 If the relaxation time of an electron is 10^{-13} sec and e/m is 10^{10} C/kg, the mobility of an electron is, (show the calculation and tick the correct one) (3)

(i) 10^{-23} kg-s/C (ii) 10^{-3} s-C/kg (iii) 10^{23} c/kg-s

Que.8 For a paramagnetic substance the ratio of magnetic energy to thermal energy at a very high magnetic is, (1)

(i) <1 (ii) >1 (iii) $=1$

Que.9 The de-Broglie wavelength for an electron of rest mass m and charge e accelerated by a potential of V volts from rest to velocity v is : (Show calculation and tick the correct one) (3)

(i) $(4eV/m)$ (ii) $(2eV/m^2)$ (iii) $(2eV/m)$ (iv) $(4eV/m^2)$